Design note: latent escalation between conflict and unrest episodes

Problem. In the current joint model, escalation is identified only at the shortest scale: κ is the effect of unrest in the previous four weeks in the same cell. Escalation over months and hundreds of kilometres is absorbed by the shared field and cannot be told apart from a shared episode. The proposal makes it identifiable.

Model. Time steps $k = 1, \ldots, K$ of Δ weeks; cells s of a coarse lattice. Each outcome has its own latent *episode field*: $a_k(s)$ for conflict and $b_k(s)$ for unrest. They evolve jointly:

$$a_k(s) = \underbrace{\phi_C\, a_{k-1}(s)}_{\text{persistence}} + \underbrace{\gamma_{U\to C}\, \bar{b}_{k-1}(s)}_{\text{escalation}} + \underbrace{\eta_k(s)}_{\text{shared shock}} + \xi_k^C(s),$$

$$b_k(s) = \phi_U\, b_{k-1}(s) + \gamma_{C\to U}\, \bar{a}_{k-1}(s) + \lambda\, \eta_k(s) + \xi_k^U(s),$$

where $\bar{b}_{k-1}(s) = (1-\omega)\, b_{k-1}(s) + \omega \cdot$ mean of b_{k-1} over the neighbours of s (so an unrest episode can spill into neighbouring cells), and likewise $\bar{a}$. The innovations η_k (shared), ξ_k^C, ξ_k^U (outcome-specific) are independent Matérn fields with sea barriers, independent over k. The observation model is unchanged: $\operatorname{logit} P(Y_{st}^C = 1) = \boldsymbol{x}^\top \beta_C + \delta_C h_C + \kappa_C h_U + \cdots + u_C + e_C + w_C + a_{k(t)}(s)$, and the same for unrest with b.

What each parameter answers.

- λ (with η): *shared episodes*, shocks that hit both outcomes in the same step and place.
- $\gamma_{U\to C}$: *latent escalation*, an unrest episode at s raising conflict risk at s and its neighbours one step later, and then decaying at rate ϕ_C. $\gamma_{C\to U}$ is the reverse direction.
- κ_C: *short-run local escalation* from recorded events, as now.

Special cases: $\gamma = 0$ gives a shared-innovation model close to the current extended model; $\gamma = \lambda = 0$ gives two independent univariate space-time models. The main summary is the impulse response: the change in conflict risk over the next $1, 2, \ldots$ steps after a one-SD unrest episode, split into its shared and its escalation parts.

Why it fits our machinery. Keeping (a, b, η) as latent variables, each step is Gaussian given the previous one, so the joint precision is sparse (block bidiagonal in time, finite-volume sparsity in space). The Laplace engine, METIS ordering and case-control subsampling are unchanged. The process is stationary if the 2×2 matrix $\begin{pmatrix} |\phi_C| & |\gamma_{U\to C}| \\ |\gamma_{C\to U}| & |\phi_U| \end{pmatrix}$ has spectral radius below one. Hyperparameters: 11 (against 9 in the extended model).

Identification. Shared episodes move both outcomes in the *same* step; escalation moves conflict *after* unrest. They are separated by timing, so the model needs enough steps: $\Delta = 4$ weeks on 1° cells for West Africa (91 steps, about 1.8×10^5 latent values) and $\Delta = 8$ on 2° cells for Africa and the Middle East (46 steps, about 1.3×10^5). Longer memory than one AR(1) can be added as a sum of a fast and a slow component, which stays sparse.

Simulation check (before any real-data claim). S2 design, 20 replicates per truth: (i) shared episodes only; (ii) latent escalation only ($\gamma_{U\to C} = 0.3$); (iii) both; (iv) short-run κ only. Fit the new model, the current extended model and the univariate model. Criteria: bias and coverage of $\gamma_{U\to C}$, λ and κ; no spurious γ under (i) and (iv). *Go/no-go:* if γ and λ are not separated in (i)–(iii), stop and fall back to regional λ and κ.

Cost. Implementation and tests: 1 day. Simulation: 1 day on the CRC. Real-data fits (two regions, warm-started, $r = 0.25$): 1–2 days. Rewriting methods and application: 1 day. Paper: Section 3 gains the latent escalation model as the main model; the current joint model becomes its $\gamma = 0$ special case; the application reports γ, λ, κ and impulse responses.